Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Microprocessor**

Code No: **CSC167**   Semester: **2nd Semester**

Full Marks: **60**   Pass Marks: **24**   Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain instruction cycle, machine cycle and T-states. Draw timing diagram of instruction cycle of STA instruction with brief description.

2. Draw block diagram of 8086 microprocessor and explain its main two functional units. Differentiate between Real Address Mode and Protected Virtual address mode of 80286 microprocessor.

3. Explain the working of PUSH and POP instruction with suitable example. Write an assembly language program using 8085 microprocessor instructions to divide 34h by 08h, store the remainder in memory location 8050h and quotient in memory location 8051h.

Section B

Attempt any Eight questions[8×5=40]

4. Differentiate between Isolated mapped and Memory Mapped I/O. Explain basic DMA operation in brief.

5. What is flag? Explain all the flags available in 8085 microprocessor.

6. Write an assembly language program for 8086 microprocessor to convert the string "microprocessor is programmable" to uppercase.

7. What is the use of AD7-AD0in 8085 microprocessor? Explain address de-multiplexing process in 8085 microprocessor with suitable diagram.

8. What is interrupt? Explain 8259 interrupt controller in brief with suitable diagram.

9. Explain Register Organization in 80386 microprocessor.

10. Draw a logic diagram showing generation of memory and I/O read/write control signals in 8085 microprocessor.

11. What is Descriptor? How logical address is converted to physical address using Descriptor table with suitable diagram.

12. Write short notes on:a)Operating Modes of 8255 PPIb) Harvard Architecture